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**MULTIMODAL VISION-PHYSIOLOGY FUSION SYSTEM FOR PREEMPTIVE
SEPSIS RISK STRATIFICATION IN INTENSIVE CARE UNITS**

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**The following specification particularly describes the invention and the manner in
which it is to be performed.**

MULTIMODAL VISION-PHYSIOLOGY FUSION SYSTEM FOR PREEMPTIVE SEPSIS RISK STRATIFICATION IN INTENSIVE CARE UNITS

FIELD OF INVENTION

The invention pertains to intelligent healthcare systems, focusing on a multimodal fusion framework that combines visual biomarkers and temporal clinical data for early detection and risk stratification of sepsis in ICU environments.

BACKGROUND OF INVENTION

In patent number US8527449B2, the approach is based on monitoring physiological parameters and generating rule-based alerts for sepsis detection; however, it lacks AI-driven prediction and does not incorporate vision-based biomarkers or multimodal data fusion. In patent number US11908581B2, the approach involves calculating sepsis risk scores using structured clinical data from monitoring devices, but it does not integrate visual inputs or advanced multimodal learning frameworks.

In patent number WO2021134027A1, the approach focuses on biomarker-based prediction of sepsis severity, yet it does not support real-time ICU monitoring or vision-based analysis.

In patent number WO2022216220A1, the approach utilizes machine learning models for sepsis detection using clinical datasets; however, it lacks multimodal fusion, visual feature extraction, and explainable AI mechanisms.

In patent number US20200005900A1, the approach applies machine learning techniques to predict patient outcomes using electronic health records, but it is not specific to sepsis and does not address ICU-based multimodal prognostication.

Accordingly, there is a need for an advanced system that integrates vision-based analysis with physiological data through a multimodal fusion framework for early and accurate sepsis prognostication.

SUMMARY OF INVENTION

The present invention relates to a multimodal vision-physiology fusion system (200) for pre-emptive sepsis risk stratification in intensive care unit patients. The invention provides an intelligent prognostic framework capable of analysing patient visual appearance together with physiological parameters, laboratory values, and temporal clinical trends for early prediction of sepsis progression. The system (200) comprises a patient monitoring camera module (201) for capturing RGB images, thermal images, infrared images, or video sequences of a patient, a clinical data acquisition module (202) for collecting physiological parameters, laboratory values, and electronic health record data, and a pre-processing module (203) for synchronization, normalization, temporal alignment, and cleaning of heterogeneous visual and clinical data streams.

The invention further comprises a neural vision model (204) comprising Vision Transformer, Convolutional Neural Network, or hybrid CNN-Transformer architectures for extraction of visual biomarkers including facial pallor, sweating pattern, respiratory movement, posture

deterioration, perfusion variation, and remote photoplethysmography-based circulatory indicators. A sequential time-series prediction module (205) comprising Long Short-Term Memory, Gated Recurrent Unit, Transformer, or Temporal Convolutional Network architectures analyses temporal physiological progression associated with sepsis onset. Visual biomarkers and temporal physiological features are processed by a multimodal fusion engine (206) employing feature concatenation layers, attention-weight assignment layers, classification layers, and sigmoid activation layers to generate a unified patient representation for sepsis prognostication.

The system (200) further comprises a sepsis risk prediction unit (207) for generation of sepsis probability scores, severity classification, and lead-time forecasting including prediction windows of 6 hours, 12 hours, and 24 hours before clinical deterioration. An explainability module (208) generates attention maps, heatmaps, temporal contribution plots, and feature-importance visualizations associated with visual and physiological parameters. A clinical display and alert generation module (209) presents risk scores, severity categories, urgency levels, and decision-support alerts through intensive care dashboards, electronic health record systems, nurse stations, or mobile notification systems. A periodic re-evaluation module (210) continuously updates patient risk assessment using newly acquired image data and clinical parameters for real-time intensive care monitoring and early clinical intervention.

The invention provides a non-contact multimodal prognostic framework that combines visual biomarkers, physiological monitoring, laboratory analysis, temporal sequence modelling, explainable artificial intelligence, and continuous reassessment into a single decision-support system. The invention enables early, continuous, and interpretable sepsis risk prediction in intensive care environments, thereby supporting timely intervention, improved patient monitoring, and enhanced clinical decision-making.

DETAIL DESCRIPTION OF THE INVENTION

The present invention relates to a vision-based multimodal prognostic system (200) for early prediction of sepsis risk in intensive care unit patients. The invention combines periodic patient image acquisition, ICU physiological monitoring, laboratory parameter analysis, electronic health record analysis, neural vision learning, sequential time-series modelling, multimodal feature fusion, explainable artificial intelligence, and clinical alert generation into a unified decision-support framework. The system (200) predicts the probability of sepsis onset within lead-time windows including 6 hours, 12 hours, and 24 hours before probable clinical onset of sepsis, thereby assisting healthcare providers in early intervention, continuous monitoring, and improved patient safety.

The system (200) comprises a patient monitoring camera module (201) positioned near an ICU patient bed. The patient monitoring camera module (201) comprises an RGB camera, thermal camera, infrared camera, depth camera, or multimodal imaging device for capturing periodic visual data of a patient without continuous physical contact. The visual data includes facial appearance, skin tone variation, body posture, respiratory movement, sweating pattern, perfusion changes, restlessness, reduced responsiveness, and other observable visual biomarkers. In one embodiment, the patient monitoring camera module (201) captures image frames at fixed intervals, while in another embodiment short-duration video sequences are

captured. The captured images and video frames are time-stamped and associated with corresponding physiological and laboratory data for simultaneous visual and clinical analysis.

A clinical data acquisition module (202) collects physiological parameters, laboratory values, and electronic health record data from ICU monitoring systems and hospital databases. The collected parameters include heart rate, respiratory rate, oxygen saturation, blood pressure, temperature, urine output, Glasgow Coma Scale values, white blood cell count, lactate concentration, C-reactive protein, procalcitonin, platelet count, creatinine, bilirubin, blood culture status, and additional laboratory or clinical indicators. The clinical data acquisition module (202) further collects patient age, sex, comorbidity profile, medication history, infection source, antibiotic administration details, ventilation status, and prior diagnosis information for generation of a comprehensive patient condition profile.

A pre-processing module (203) processes raw visual and clinical data before machine learning analysis. The pre-processing module (203) removes noisy image frames, corrects illumination variation, resizes image frames, normalizes pixel values, calibrates thermal distributions, synchronizes video frames with ICU monitor values using timestamps, handles missing clinical values, removes outliers, normalizes numerical parameters, encodes categorical variables, and aligns heterogeneous data streams into uniform temporal windows. The pre-processing module (203) addresses irregularity, incompleteness, and noise commonly associated with ICU datasets.

A neural vision model (204) extracts visual biomarkers from patient images and video sequences. The neural vision model (204) comprises a Vision Transformer architecture, Convolutional Neural Network architecture, or hybrid CNN-Transformer architecture for analysis of facial pallor, skin mottling, and sweating, peripheral perfusion, respiratory distress, reduced movement, altered facial expression, posture deterioration, respiratory rhythm, and motion-related abnormalities. In one embodiment, frame-level feature extraction is performed from static images, while in another embodiment motion-based features are extracted from temporal frame sequences.

The neural vision model (204) further performs remote photoplethysmography-based visual analysis using subtle colour variations in facial or visible skin regions. The remote photoplethysmography analysis derives non-contact pulse-related features, perfusion trends, heart-rate variability indicators, and circulatory instability patterns associated with sepsis progression. The visual stream therefore acts as an additional biomarker source for early deterioration detection.

A visual biomarker extraction unit associated with the neural vision model (204) converts image information into clinically meaningful features including facial colour variation, skin temperature distribution, capillary perfusion-related patterns, respiratory movement amplitude, sweating intensity, patient motion score, posture change frequency, facial expression score, and attention-based visual abnormality regions. These features are combined with physiological and laboratory parameters for generation of a richer patient representation.

A clinical feature extraction subsystem processes structured ICU data and extracts temporal physiological features including heart-rate trends, respiratory-rate progression, oxygen saturation decline, abnormal temperature patterns, rising lactate levels, increasing white blood cell count, decreasing platelet count, worsening renal parameters, and indicators associated with infection and organ dysfunction. The clinical feature extraction subsystem studies temporal direction, progression rate, and deterioration pattern of physiological parameters instead of isolated measurements.

A sequential time-series prediction module (205) processes clinical data over time using Long Short-Term Memory networks, Gated Recurrent Units, Temporal Convolutional Networks, Transformer architectures, or hybrid temporal learning systems. The sequential time-series prediction module (205) learns progressive physiological patterns preceding sepsis onset and generates temporal clinical feature vectors representing current and future patient risk states.

A multimodal fusion engine (206) combines visual features generated by the neural vision model (204) with clinical features generated by the sequential time-series prediction module (205). In one embodiment, feature concatenation followed by fully connected classification layers and sigmoid activation layers generates unified patient representations. In another embodiment, the multimodal fusion engine (206) employs attention mechanisms assigning different importance weights to visual, physiological, and laboratory features according to their contribution toward predicted sepsis risk. The multimodal fusion engine (206) therefore determines whether visual deterioration, physiological instability, or laboratory abnormality contributes more strongly toward predicted clinical deterioration.

The multimodal fusion engine (206) generates a unified patient feature representation comprising visible patient condition and internal physiological status. The unified representation reduces dependence upon any single data source. When laboratory values become delayed, visual and physiological streams continue contributing toward risk estimation. Similarly, when camera visibility becomes unclear, temporal clinical modelling continues supporting prediction.

A sepsis risk prediction unit (207) receives the unified patient representation and estimates the probability of sepsis onset. The sepsis risk prediction unit (207) generates probability scores between zero and one and classifies patient condition into low-risk, moderate-risk, high-risk, or critical-risk categories. The sepsis risk prediction unit (207) additionally forecasts sepsis progression at lead-time intervals including 6 hours, 12 hours, and 24 hours before clinical diagnosis.

The sepsis risk prediction unit (207) applies decision thresholds to predicted probability scores for generation of clinical alerts. In one embodiment, fixed thresholds are employed. In another embodiment, personalized thresholds are selected according to patient age, comorbidity profile, baseline vital signs, and severity scores.

An explainability module (208) generates attention maps, heatmaps, feature-importance scores, saliency visualizations, and temporal contribution plots associated with visual and

clinical parameters. Attention maps highlight influential image regions including face, skin, chest movement, or peripheral regions, while temporal contribution plots identify influential physiological parameters including lactate progression, respiratory instability, blood pressure decline, and oxygen saturation reduction. The explainability module (208) therefore improves interpretability and clinical trust associated with artificial intelligence predictions.

A clinical display and alert generation module (209) presents predicted risk scores, severity categories, attention maps, temporal contribution plots, urgency levels, and clinical recommendations through ICU dashboards, nurse stations, electronic health record systems, bedside displays, or mobile alert systems. Alert notifications may include warnings such as “High risk of sepsis within 12 hours” together with contributing visual and physiological features. The clinical display and alert generation module (209) further produces alarms and notification messages when patients cross high-risk thresholds.

A periodic re-evaluation module (210) continuously updates sepsis risk predictions as new visual and clinical data becomes available. The periodic re-evaluation module (210) continuously acquires updated image streams and physiological parameters, recalculates extracted features, refreshes multimodal fusion outputs, and updates probability scores for real-time ICU monitoring. The periodic re-evaluation module (210) further supports severity classification including mild risk, moderate risk, high risk, critical risk, septic shock progression, and organ dysfunction forecasting.

A major inventive feature of the present invention involves combining non-contact visual biomarkers with structured ICU time-series analysis. Existing systems generally depend upon vital signs, laboratory values, scoring systems, or electronic health records alone. The present invention introduces patient visual appearance as an additional prognostic source. The use of Vision Transformer or Convolutional Neural Network architectures for ICU image analysis together with sequential physiological modelling creates a multimodal prognostic framework capable of identifying deterioration patterns earlier than conventional systems.

Another inventive feature involves fusion of remote photoplethysmography indicators, perfusion variation, respiratory movement, facial appearance, and physiological parameters into a unified patient representation. The system (200) therefore studies visual and physiological deterioration simultaneously instead of analysing isolated abnormalities independently.

The invention further provides multi-lead-time prediction before clinical confirmation of sepsis. Forecasting six hours ahead supports urgent intervention, twelve-hour prediction supports clinical review and treatment planning, and twenty-four-hour prediction supports preventive care planning and early risk stratification.

The invention additionally improves clinical usability through explainable artificial intelligence outputs. Attention maps and feature contribution values assist clinicians in understanding prediction rationale and verifying clinical consistency, thereby reducing black-box behaviour associated with conventional artificial intelligence systems.

During operation, the patient monitoring camera module (201) captures patient images while the clinical data acquisition module (202) collects physiological and laboratory information. The pre-processing module (203) cleans and synchronizes heterogeneous datasets before transmission toward the neural vision model (204) and sequential time-series prediction module (205). Extracted visual and clinical features are processed by the multimodal fusion engine (206), after which the sepsis risk prediction unit (207) estimates probability of sepsis onset and determines patient risk category. The explainability module (208) generates interpretable visualizations, while the clinical display and alert generation module (209) communicates alerts and decision-support information to healthcare providers. The periodic re-evaluation module (210) continuously refreshes prediction outputs using newly acquired data.

Thus, the present invention provides a comprehensive ICU decision-support system combining camera-based patient observation, remote visual biomarker extraction, physiological monitoring, laboratory analysis, temporal sequence modelling, multimodal feature fusion, explainable artificial intelligence, and clinical alert generation for early, continuous, and interpretable sepsis prognostication in intensive care environments.

BRIEF DESCRIPTION OF DRAWINGS

Figure 1 illustrates the overall architecture of the proposed vision-based prognostic system for early prediction of sepsis in ICU patients. The system comprises multiple functional modules including data acquisition, pre-processing, feature extraction, multimodal fusion, and prediction. The data acquisition unit collects both visual data through patient monitoring cameras and physiological data such as vital signs and laboratory values from ICU devices.

The pre-processing module performs normalization, noise reduction, and synchronization of heterogeneous data streams. The feature extraction module derives visual biomarkers using vision models and clinical features from physiological data. These features are combined using a multimodal fusion mechanism to generate a unified patient representation. The prediction and prognostication engine processes the fused data to estimate sepsis risk and supports early clinical decision-making through a feedback loop.

Figure 2 illustrates a flowchart representing the workflow of the vision-based prognostic system for early sepsis prediction. The process begins with periodic capture and analysis of patient images using vision-based models such as Vision Transformer (ViT) or Convolutional Neural Networks (CNN), alongside continuous collection of ICU physiological data processed using sequential models such as Transformer or Long Short-Term Memory (LSTM) networks. The extracted visual and clinical features are fused using a multimodal fusion model. Based on the computed sepsis risk probability, the system determines whether the risk exceeds a predefined threshold. If the threshold is met, alerts are generated and healthcare providers are notified along with visualization of attention maps and risk scores. Otherwise, the system performs periodic re-evaluation and updates the data, ensuring continuous monitoring and timely intervention.

Figure 3 illustrates a detailed block diagram of the vision-based prognostic system showing the interaction between various functional components. A patient monitoring camera captures periodic thermal or RGB images, which are processed by a neural vision model such as ViT or CNN to extract visual indicators. Simultaneously, a clinical data acquisition unit gathers physiological parameters, laboratory values, and vital signs, which are processed by a

sequential time-series prediction module using Transformer or LSTM architectures. The visual and clinical features are integrated through a multimodal fusion engine employing feature concatenation and classification layers. The fused representation is then passed to a sepsis risk prediction unit that computes the probability of sepsis onset, including early prediction horizons such as 6, 12, or 24 hours in advance. An explainability module generates attention maps and heatmaps to interpret model decisions. Finally, the output prediction unit communicates results through a clinical display interface, enabling visualization of risk scores, alerts, and decision-support information for healthcare providers.

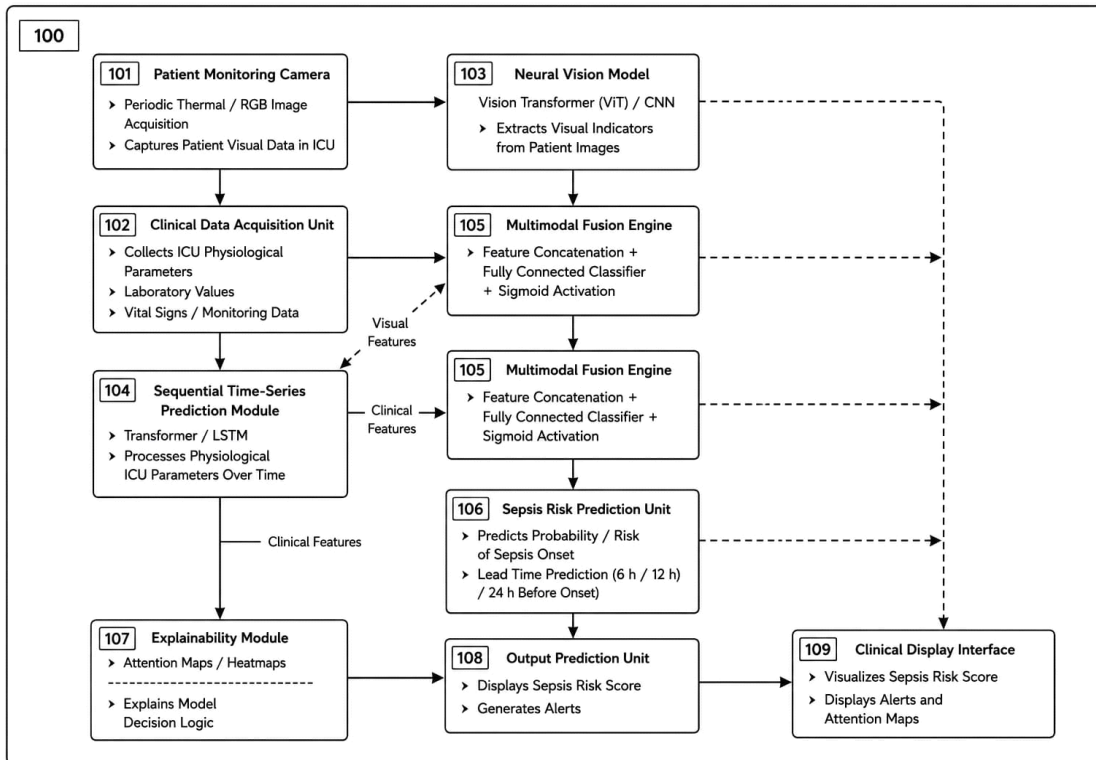


Figure 1

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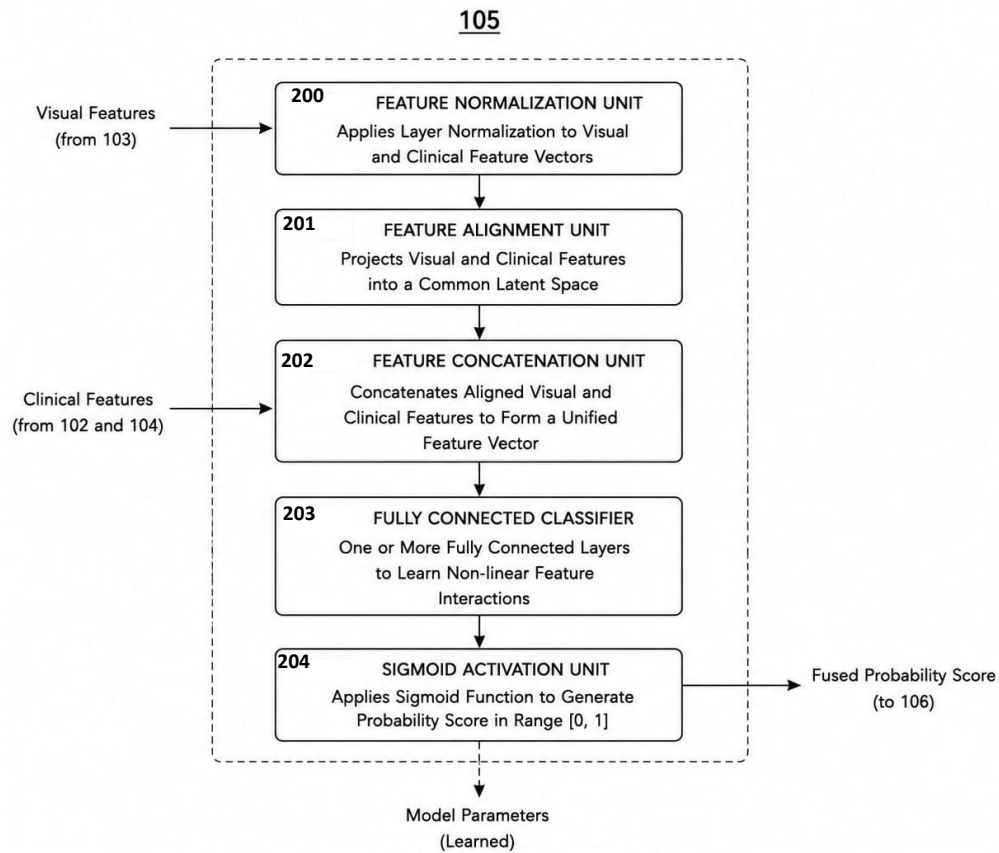


Figure 2

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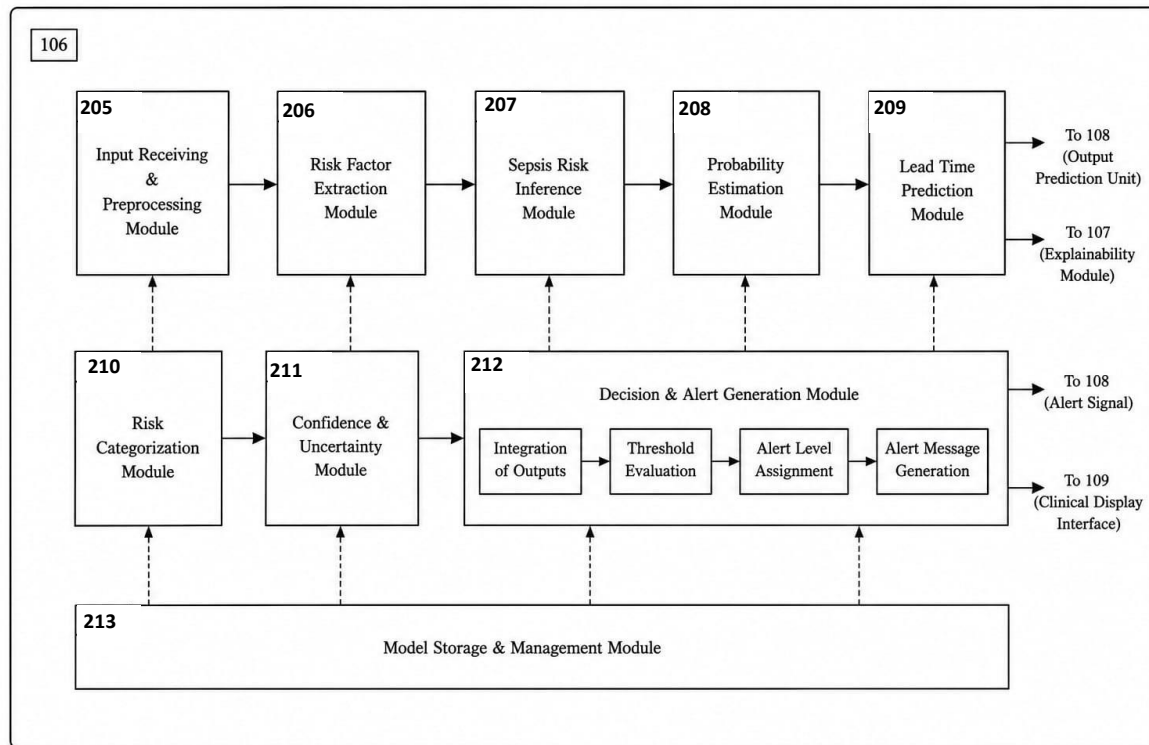


Figure 3

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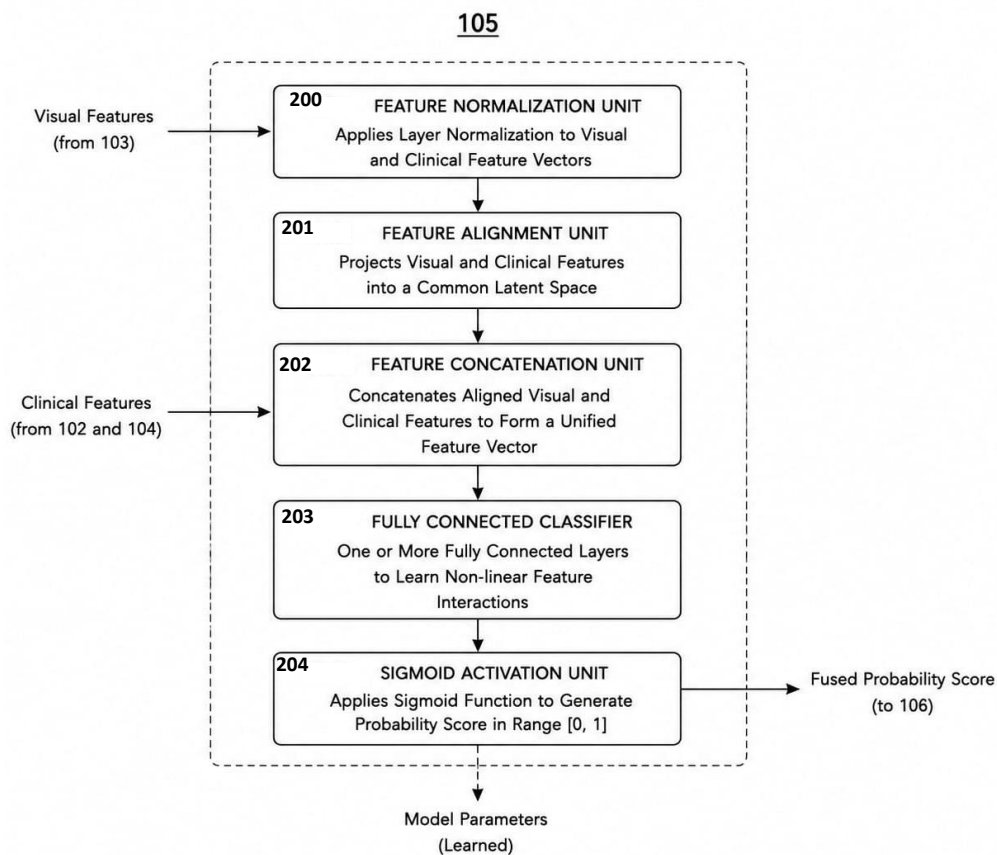
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MULTIMODAL VISION-PHYSIOLOGY FUSION SYSTEM FOR PREEMPTIVE SEPSIS RISK STRATIFICATION IN INTENSIVE CARE UNITS

ABSTRACT OF THE INVENTION

The present invention relates to a multimodal vision-physiology fusion system (200) for early prediction and stratification of sepsis risk in intensive care unit patients. The system (200) comprises a patient monitoring camera module (201) for capturing RGB, thermal, infrared, or video-based patient data, a clinical data acquisition module (202) for collecting physiological parameters, laboratory values, and electronic health record data, and a pre-processing module (203) for synchronization and normalization of heterogeneous datasets. A neural vision model (204) extracts visual biomarkers including facial pallor, respiratory movement, perfusion variation, and remote photoplethysmography indicators, while a sequential time-series prediction module (205) analyses temporal physiological progression associated with sepsis onset. A multimodal fusion engine (206) combines visual and clinical features to generate unified patient representations for sepsis prognostication. A sepsis risk prediction unit (207) generates probability scores, severity classification, lead-time forecasting, explainable visualizations, and clinical alerts for continuous intensive care monitoring and early intervention.



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WE CLAIM

1. A system (100) for early prediction of sepsis in intensive care unit patients, comprising:
 - a. A patient monitoring camera (101) to capture thermal images, RGB images, infrared images, or video frames of a patient;
 - b. A clinical data acquisition unit (102) to obtain physiological parameters, laboratory parameters, and electronic health record data from intensive care monitoring devices; and
 - c. A processing unit to determine a probability of sepsis onset using acquired visual data and clinical data.
2. The system (100) as claimed in claim 1, wherein the processing unit comprises:
 - a. A neural vision model (103) comprising a Vision Transformer, Convolutional Neural Network, or hybrid CNN-Transformer architecture for extraction of visual biomarkers including facial pallor, skin perfusion variation, respiratory movement, sweating pattern, posture variation, restlessness, and remote photoplethysmography signals; and
 - b. A sequential time-series prediction module (104) comprising a Transformer, Long Short-Term Memory network, Gated Recurrent Unit, or Temporal Convolutional Network for extraction of temporal physiological features from continuously varying clinical parameters.
3. The system (100) as claimed in claim 1, wherein visual biomarkers from the neural vision model (103) and temporal physiological features from the sequential time-series prediction module (104) are processed by a multimodal fusion engine (105) comprising feature concatenation layers, attention-weight assignment layers, classification layers, and sigmoid activation layers to generate a unified patient representation for sepsis prognostication.
4. The system (100) as claimed in claim 1, further comprising a sepsis risk prediction unit (106) to generate a sepsis probability score, classify patient condition into low-risk, moderate-risk, high-risk, or critical-risk categories, and forecast sepsis onset at lead-time intervals comprising 6 hours, 12 hours, and 24 hours before clinical deterioration.
5. The system (100) as claimed in claim 1, further comprising an explainability module (107) to generate attention maps, heatmaps, feature contribution plots, and temporal importance visualizations associated with visual biomarkers and physiological parameters, and a clinical display interface (108) to present sepsis risk scores, alerts, urgency levels, and decision-support outputs to healthcare personnel.
6. The system (100) as claimed in claim 1, wherein the patient monitoring camera (101) derives non-contact physiological indicators using remote photoplethysmography analysis from facial or skin-region colour variations to estimate pulse characteristics, perfusion trends, circulatory instability patterns, and heart-rate variability indicators associated with sepsis progression.
7. The system (100) as claimed in claim 1, further comprising a periodic re-evaluation module (109) to continuously update patient risk assessment using newly acquired image data and clinical parameters, revise prediction outputs according to temporal patient progression, and support continuous intensive care monitoring.

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